Spring of Code

Artificial Intelligence

(Machine Learning & Deep Learning)

Instructor
Asad Ullah
- *Research Assistant* (DIP Lab)

Duration
03 Months
(Feb – April)

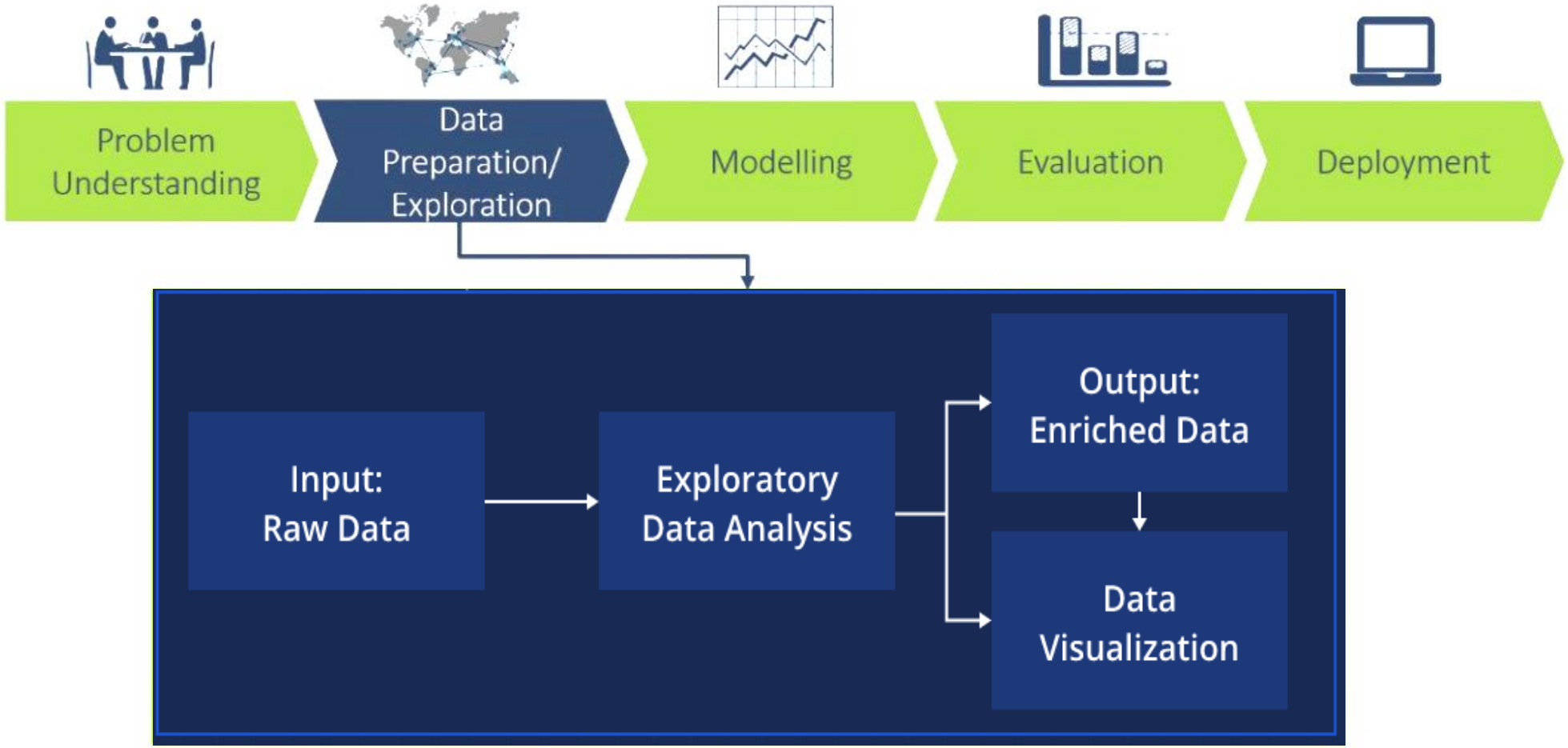
Day 01 – Exploratory Data Analysis
(Introduction to NumPy)

Objectives:

- What is Exploratory Data Analysis?
- Introduction to NumPy
- NumPy Arrays

Exploratory Data Analysis

The process of examining datasets to summarize their main characteristics. It is a crucial step in the data analysis workflow to gain a deep understanding of the dataset before modeling.



Exploratory Data Analysis

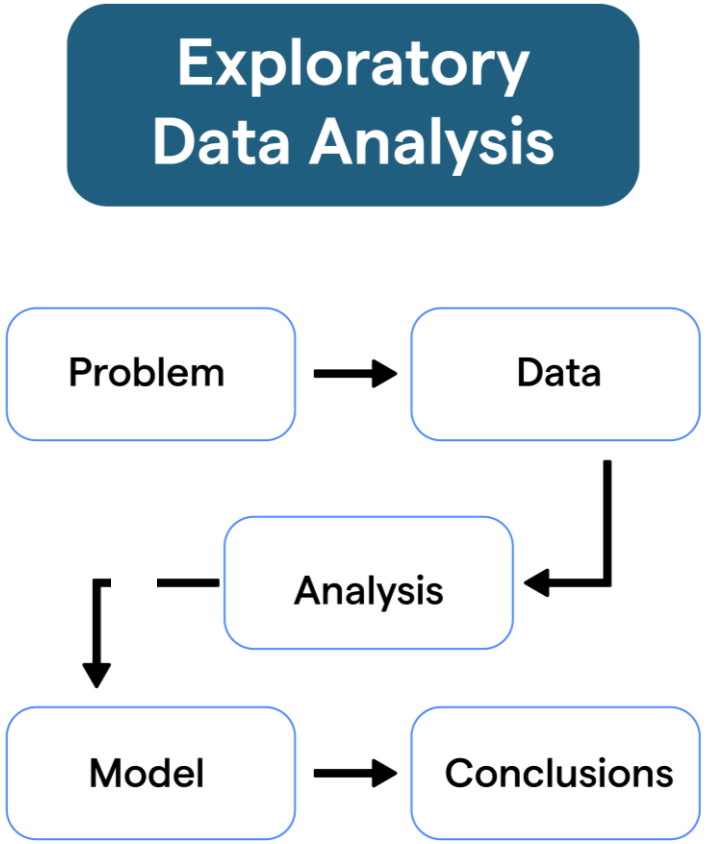
- The process of examining datasets – often with visual methods – to summarize their main characteristics.
- It is a crucial step in the data analysis workflow to gain a deep understanding of the dataset before modeling.

Objectives:

- Understand data structure and underlying patterns.
- Identify anomalies, missing values, and outliers.
- Detect trends and relationships between variables.
- Form hypothesis to inform further analysis or modeling.

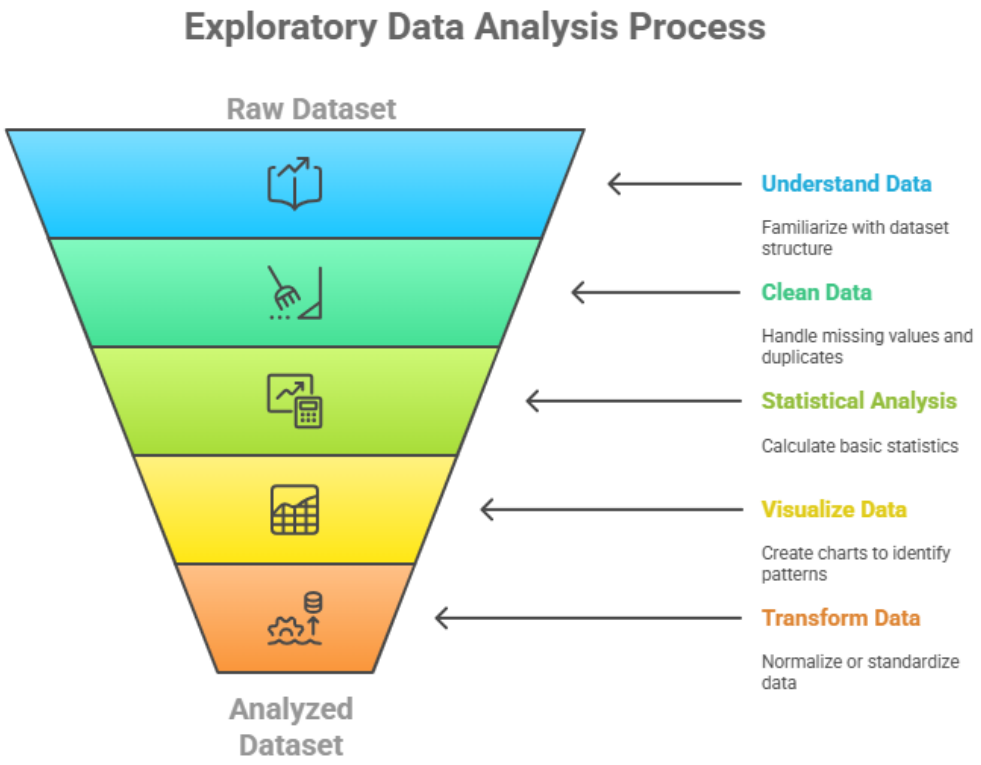
Importance:

- Provides insights for data-driven decision making.
- Improves predictive model quality by identifying issues early.
- Ensures data integrity and readiness for analysis.



Key Steps in EDA

- **Understanding the Data:** Get familiar with the dataset, check number of rows, columns, and data types.
- **Data Cleaning:** Handle missing values, duplicates, and inconsistencies.
- **Statistical Analysis:** Use basic statistics (mean, median, standard deviation) to summarize each variable.
- **Data Visualization:** Use charts to uncover patterns, trends and outliers.
- **Data Transformation** (if needed)**:** Normalize or standardize values, or convert data into a better format for further analysis or modeling.



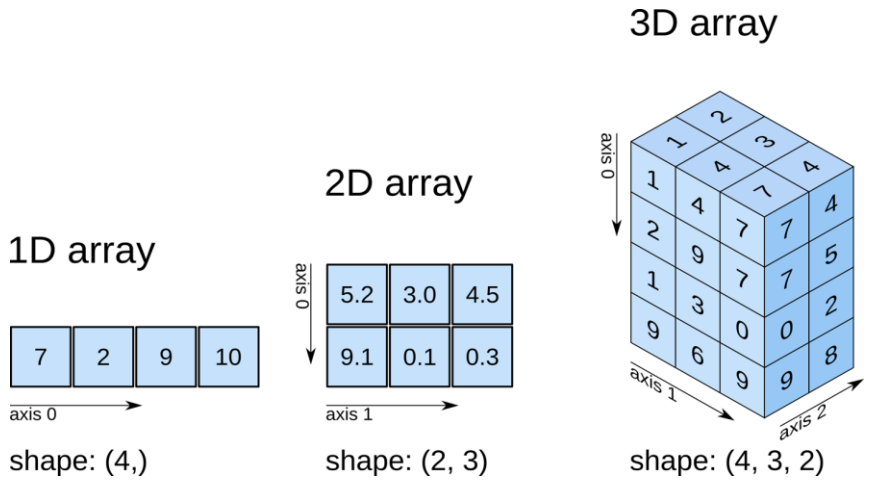
Python Libraries for EDA

- **`NumPy:`** Essential for numerical operations in Python, it provides support for multi-dimensional arrays, along with mathematical functions on these arrays.
- **`Pandas:`** Library for data manipulation and analysis. It makes it easy to clean, transform, and aggregate data.
- **`Matplotlib:`** A versatile plotting library used to create static, interactive, and animated visualizations in Python.
- **`sklearn:`** Primarily a machine learning library but includes many tools useful for data preprocessing and feature selection, which are key parts of EDA.

Introduction to NumPy

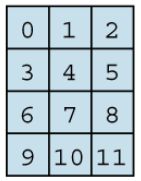
NumPy: "Numerical Python" – The foundation of scientific computing in Python

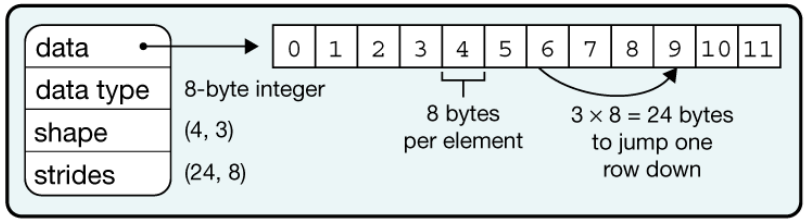
- **Initial release:** As **Numeric** (1995); as NumPy, (2006)
- Python library for efficient array operations and mathematical computations
- **Core Data Structure**: N-dimensional array (ndarray)



Why NumPy?

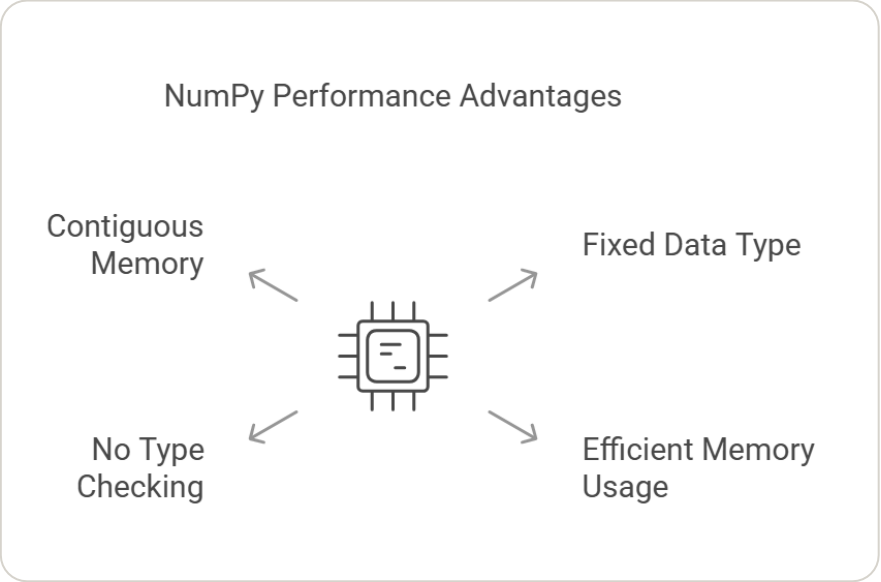
- **Speed:** Up to 50x faster than Python lists.
- **Memory Efficient:** Uses less memory than traditional Python data structures.
- **Foundation:** Backend for Pandas, SciPy, Matplotlib, and machine learning libraries
- **Versatile:** Supports 1D arrays, matrices, tensors, and higher dimensions.





Why NumPy Arrays are Faster Than Lists

1. **Fixed Data Type**:
   - NumPy arrays have a uniform data type, which eliminates the overhead of managing different data types like in Python lists.
2. **Contiguous Memory Storage**:
   - Data stored in continuous memory blocks enhances cache efficiency and reduces memory consumption.
3. **No Runtime Type Checking**:
   - Operations on NumPy arrays skip runtime type checks, making computations faster compared to Python lists.
4. **Optimized Implementation**:
   - Core operations written in C and Fortran for maximum performance.



Differences Between Python Lists and NumPy Arrays:

Feature	Python Lists	NumPy Arrays
Data Type	Mixed Types	Homogeneous types
Memory Efficiency	Low	High
Computation Speed	Slow	Fast

Thank You